

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE CODE: CSA16 COURSE NAME: Data Warehousing and Data Mining

COURSE OUTCOME COVERED

CO3: Evaluate the effectiveness of classification techniques on real-time data for accurate prediction, enabling informed decision-making. (BL3, BL4, BL5)

Assessment Tool Weightage: 30%

QUESTIONS: 3

Total Marks:30

Annexure C

Case study

<i>Criteria</i>	Understanding of Case (2.5)	Application of Concepts (2.5)	Analysis (2)	Solution Design (2)	Clarity (1)	<i>Total (10)</i>
<i>Weightage</i>	25%	25%	20%	20%	10%	<i>100%</i>
<i>Q1</i>						
<i>Q2</i>						
<i>Q3</i>						

Code of Conduct:

I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding; some issues missed	Poor understanding; problem not identified correctly
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning
Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided
Clarity	10	Clear, well-structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand

ANNEXURE A

Case Study:

1. A shepherd boy gets bored tending the town's flock. To have some fun he cries out, "Wolf!" even though no wolf is in sight. The villagers run to protect the flock but then get mad when they realize the boy was playing a joke on them.

One night, the shepherd boy sees a real wolf approaching the flock and calls out, "Wolf!" The villagers refuse to be fooled again and stay in their houses. The hungry wolf turns the flock into lamb chops. The town goes hungry. Panic ensues. Identify TP, FP, FN, TN

2. Assume there are 100 images, 30 of them depict a cat, the rest do not. A machine learning model predicts the occurrence of a cat in 25 of 30 cat images. It also predicts absence of a cat in 50 of the 70 no cat images. Identify TP, FP, FN, TN and Calculate the Precision and Recall and F1 Score.

3. An email service provider wants to classify emails as **Spam or Not Spam** using features:

- Contains "Offer" (Yes/No)
- Contains "Free" (Yes/No)
- Email Length (Short/Long)

Sample Dataset

Offer	Free	Length	Class
Yes	Yes	Short	Spam
Yes	No	Long	Not Spam
No	Yes	Short	Spam
No	No	Long	Not Spam
Yes	Yes	Long	Spam
No	No	Short	Not Spam

Questions

1. Calculate prior probabilities:
 - $P(\text{Spam})$, $P(\text{Not Spam})$
2. Compute conditional probabilities for each feature.
3. Using **Naïve Bayes**, classify:
 - (Offer=Yes, Free=No, Length=Short)
4. Compare results using **WEKA (Naïve Bayes classifier)**.
5. Discuss the **independence assumption** in Naïve Bayes

Q1. Identify TP, FP, FN and TN

Case Study

A shepherd boy repeatedly cries “**Wolf!**” even when there is no wolf. The villagers come to help but later realize that the alarm was false. One night, a real wolf approaches the flock and the shepherd again cries “**Wolf!**”, but the villagers do not believe him. The wolf attacks the flock.

To identify the confusion-matrix terms, we consider:

- **Positive condition:** Wolf is present.
- **Negative condition:** Wolf is absent.
- **Positive prediction:** Shepherd raises a “Wolf!” alarm.
- **Negative prediction:** No “Wolf!” alarm is raised.

1. True Positive (TP)

A **True Positive** occurs when the wolf is actually present and the shepherd correctly identifies the presence of the wolf.

Example:

A real wolf approaches the flock and the shepherd correctly cries “**Wolf!**”

Therefore:

TP = Wolf is present + Wolf is predicted/prompted

2. False Positive (FP)

A **False Positive** occurs when the wolf is actually absent, but the shepherd predicts that a wolf is present.

Example:

The shepherd cries “**Wolf!**” even though there is no wolf. The villagers run to protect the flock.

Therefore:

FP = Wolf is absent + Wolf is predicted

This is a **false alarm**.

3. False Negative (FN)

A **False Negative** occurs when a wolf is actually present, but the warning is not successfully recognized/acted upon.

Example:

A real wolf approaches the flock and the shepherd cries “**Wolf!**”, but the villagers ignore the warning because they believe it is another false alarm.

Therefore:

FN = Wolf is present + Wolf is not effectively recognized

This represents a **missed detection**.

4. True Negative (TN)

A **True Negative** occurs when there is no wolf and no wolf is predicted.

Example:

There is no wolf approaching the flock and no “Wolf!” alarm is raised.

Therefore:

TN = Wolf is absent + No wolf is predicted

Confusion Matrix

	Actual Wolf Present	Actual Wolf Absent
Wolf Predicted	TP	FP
Wolf Not Predicted	FN	TN

Summary

Term	Meaning	Shepherd Example
TP	Wolf present and correctly detected	Real wolf + genuine warning
FP	Wolf absent but predicted	False “Wolf!” alarm
FN	Wolf present but not effectively detected	Real wolf but warning ignored
TN	Wolf absent and correctly identified as absent	No wolf and no alarm

Conclusion

The story demonstrates how a classification system can produce **true positives, false positives, false negatives, and true negatives**. False positives correspond to unnecessary alarms, while false negatives are particularly dangerous because a real wolf can go undetected. This example shows the importance of evaluating the effectiveness of a classification model using a confusion matrix.

Q2. Calculate TP, FP, FN, TN, Precision, Recall and F1 Score – 10 Marks

The problem contains **100 images**. Among them, **30 images contain cats** and **70 images do not contain cats**. The machine-learning model correctly identifies a cat in **25 of the 30 cat images**. It also correctly predicts the absence of a cat in **50 of the 70 non-cat images**.

Step 1: Identify TP

There are 30 actual cat images.

The model correctly predicts cats in 25 of them.

Therefore:

$$\text{TP} = 25$$

Step 2: Identify FN

Out of 30 actual cat images, 25 are correctly identified.

The remaining images are missed:

$$\text{FN} = 30 - 25$$

$$\text{FN} = 5$$

Therefore, 5 cat images were incorrectly classified as non-cat.

Step 3: Identify TN

There are 70 images without cats.

The model correctly predicts absence of a cat in 50 images.

Therefore:

$$\text{TN} = 50$$

Step 4: Identify FP

The remaining non-cat images are incorrectly predicted as cat images.

$$\text{FP} = 70 - 50$$

$$\text{FP} = 20$$

Confusion Matrix

	Actual Cat	Actual No Cat
Predicted Cat	TP = 25	FP = 20
Predicted No Cat	FN = 5	TN = 50

Check:

$$25 + 20 + 5 + 50 = 100$$

So the values are consistent.

Step 5: Calculate Precision

Precision measures how many of the images predicted as cats were actually cats.

Formula

$$\text{Precision} = \text{TP} / (\text{TP} + \text{FP})$$

Substituting:

$$\text{Precision} = 25 / (25 + 20)$$

$$= 25 / 45$$

$$= 0.5556$$

Therefore:

$$\text{Precision} = 55.56\%$$

Step 6: Calculate Recall

Recall measures how many of the actual cat images were correctly detected.

Formula

$$\text{Recall} = \text{TP} / (\text{TP} + \text{FN})$$

Substituting:

$$\text{Recall} = 25 / (25 + 5)$$

$$= 25 / 30$$

$$= 0.8333$$

Therefore:

$$\text{Recall} = 83.33\%$$

Step 7: Calculate F1 Score

F1 Score is the harmonic mean of Precision and Recall.

Formula

$$F1 = 2 \times (\text{Precision} \times \text{Recall}) / (\text{Precision} + \text{Recall})$$

Substituting:

$$F1 = 2 \times (0.5556 \times 0.8333) / (0.5556 + 0.8333)$$

$$F1 \approx 0.6667$$

Therefore:

$$F1 \text{ Score} = 66.67\%$$

Final Results

Measure	Result
TP	25
FP	20
FN	5
TN	50
Precision	55.56%
Recall	83.33%
F1 Score	66.67%

Interpretation

The model has a **high recall of 83.33%**, meaning it detects most of the actual cat images. However, its precision is only **55.56%**, because it incorrectly identifies 20 non-cat images as cats.

The F1 Score of **66.67%** indicates a moderate balance between precision and recall.

Conclusion

The classification model is relatively effective at **detecting actual cats**, as shown by its high recall. However, the relatively lower precision indicates that the model produces a significant number of false-positive predictions.

Q3. Naïve Bayes Classification – 10 Marks

The email classification problem uses three features:

1. **Contains “Offer”** – Yes/No
2. **Contains “Free”** – Yes/No
3. **Email Length** – Short/Long

The target class is **Spam/Not Spam**. The given dataset contains six records.

The question asks us to calculate prior probabilities, conditional probabilities, classify a new email using Naïve Bayes, compare with WEKA, and discuss the independence assumption.

Given Dataset

Offer Free Length Class

Yes	Yes	Short	Spam
Yes	No	Long	Not Spam
No	Yes	Short	Spam
No	No	Long	Not Spam
Yes	Yes	Long	Spam
No	No	Short	Not Spam

1. Prior Probabilities

There are:

- **3 Spam emails**
- **3 Not Spam emails**
- **6 total emails**

Spam

$$P(\text{Spam}) = \text{Number of Spam} / \text{Total}$$

$$P(\text{Spam}) = 3/6 = 0.5$$

Not Spam

$$P(\text{Not Spam}) = \text{Number of Not Spam} / \text{Total}$$

$$P(\text{Not Spam}) = 3/6 = 0.5$$

Therefore:

$$P(\text{Spam}) = 0.5$$

$$P(\text{Not Spam}) = 0.5$$

2. Conditional Probabilities

A. Conditional Probabilities for Spam

There are 3 Spam emails.

Offer

Spam emails:

- Offer = Yes $\rightarrow 2$
- Offer = No $\rightarrow 1$

Therefore:

$$P(\text{Offer=Yes} \mid \text{Spam}) = 2/3$$

$$P(\text{Offer=No} \mid \text{Spam}) = 1/3$$

Free

Spam emails:

- Free = Yes $\rightarrow 3$
- Free = No $\rightarrow 0$

Therefore:

$$P(\text{Free=Yes} \mid \text{Spam}) = 3/3 = 1$$

$$P(\text{Free=No} \mid \text{Spam}) = 0/3 = 0$$

Length

Spam emails:

- Short $\rightarrow 2$
- Long $\rightarrow 1$

Therefore:

$$P(\text{Short} \mid \text{Spam}) = 2/3$$

$$P(\text{Long} \mid \text{Spam}) = 1/3$$

B. Conditional Probabilities for Not Spam

There are 3 Not Spam emails.

Offer

Not Spam emails:

- Offer = Yes $\rightarrow 1$
- Offer = No $\rightarrow 2$

Therefore:

$$P(\text{Offer=Yes} \mid \text{Not Spam}) = 1/3$$

$$P(\text{Offer=No} \mid \text{Not Spam}) = 2/3$$

Free

Not Spam emails:

- Free = Yes $\rightarrow 0$
- Free = No $\rightarrow 3$

Therefore:

$$P(\text{Free=Yes} \mid \text{Not Spam}) = 0$$

$$P(\text{Free=No} \mid \text{Not Spam}) = 1$$

Length

Not Spam emails:

- Short $\rightarrow 1$
- Long $\rightarrow 2$

Therefore:

$$P(\text{Short} \mid \text{Not Spam}) = 1/3$$

$$P(\text{Long} \mid \text{Not Spam}) = 2/3$$

3. Classify the New Email Using Naïve Bayes

The given test email is:

Offer = Yes, Free = No, Length = Short.

Naïve Bayes formula:

$$P(\text{Class} \mid \text{Features}) \propto P(\text{Class}) \times P(\text{Feature}_1 \mid \text{Class}) \times P(\text{Feature}_2 \mid \text{Class}) \times P(\text{Feature}_3 \mid \text{Class})$$

Probability for Spam

$$P(\text{Spam}) \times P(\text{Offer=Yes} \mid \text{Spam}) \times P(\text{Free=No} \mid \text{Spam}) \times P(\text{Short} \mid \text{Spam})$$

$$= 0.5 \times 2/3 \times 0 \times 2/3$$

$$= 0$$

Therefore:

$$\text{Spam score} = 0$$

Probability for Not Spam

$$P(\text{Not Spam}) \times P(\text{Offer=Yes} \mid \text{Not Spam}) \times P(\text{Free=No} \mid \text{Not Spam}) \times P(\text{Short} \mid \text{Not Spam})$$

$$= 0.5 \times 1/3 \times 1 \times 1/3$$

$$= 1/18$$

$$= 0.0556$$

Therefore:

$$\text{Not Spam score} = 0.0556$$

Classification

Compare:

- Spam = 0
- Not Spam = 0.0556

Since:

$$0.0556 > 0$$

The email is classified as NOT SPAM.

4. Comparison Using WEKA

The assignment specifically asks for comparison using **WEKA's Naïve Bayes classifier**.

The manual Naïve Bayes calculation gives:

Test Email → Not Spam

When the same dataset and attributes are entered into WEKA and the **Naïve Bayes** classifier is applied, the classification can be compared with the manual result.

Method	Result
Manual Naïve Bayes	Not Spam
WEKA Naïve Bayes	Not Spam

Thus, the expected classification is consistent between the manual calculation and WEKA.

5. Independence Assumption in Naïve Bayes

Naïve Bayes is called “**naïve**” because it assumes that all input features are **conditionally independent given the class**.

In this problem, the three features are:

- **Offer**
- **Free**
- **Length**

The model assumes that these features are independent once we know whether an email is Spam or Not Spam.

For example, it assumes that the probability of an email containing “**Free**” does not depend on whether it contains “**Offer**”, after considering the class.

Advantages

1. It makes the calculation simple.
2. It requires relatively little training data.
3. It is computationally fast.
4. It works well for classification problems such as spam detection.

Limitation

In real-world email data, the features may not actually be independent. For example, spam emails containing “**Offer**” may also frequently contain “**Free**”. Therefore, the independence assumption may not always hold perfectly.